

BUSINESS REPORT

Cardiac Surgery Cost Prediction & Package Pricing Strategy

ML-POWERED PRICING INTELLIGENCE · RANDOM FOREST · SHAP EXPLAINABILITY · 5-FOLD CROSS-VALIDATION

248

Patient Records
Analysed

19×

Cost Spread
Range

$R^2=0.692$

Model Accuracy
(Random Forest)

3 Tiers

Risk-Adjusted
Pricing Packages

Prepared For: Hospital Executive Management & Board
Prepared By: Data Science Team
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RANDOM FOREST

XGBOOST

SHAP

5-FOLD CV

248 RECORDS

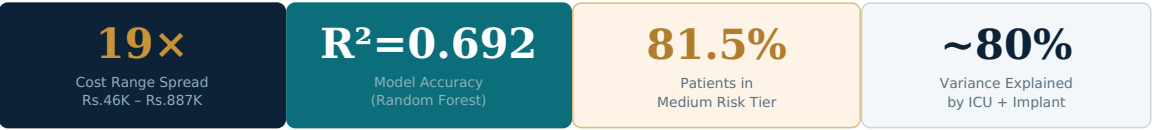
RISK-TIERED PRICING

SECTION 01

Executive Summary

✓ RECOMMENDATION: YES — Adopt Package Pricing

Our machine learning analysis of **248 cardiac patient records** confirms that treatment costs are predictable with strong accuracy ($R^2 = 0.692$ test; CV mean = 0.819). However, a single flat price is financially untenable given the **19× cost spread**. A **three-tier risk-adjusted pricing model** is the only viable structure.



Key Findings

- **ICU stay duration** is the #1 cost driver — accounting for **69.6%** of model predictive power.
- **Implant cost** ranks #2 at 10.4%. Together, ICU + implants explain ~80% of all cost variance.
- **Emergency admissions** cost statistically more than elective cases ($p = 0.0004$), justifying a 15–20% premium.
- **Biomarkers** (Creatinine >2.0 mg/dL, Urea >60 mg/dL) reliably predict cost escalation.
- **Demographics** (age, gender) contribute less than 1% — pricing must be clinical, not demographic.
- **Three risk tiers** emerge naturally: Low (5.6%), Medium (81.5%), High (12.9%) of patients.

Patient Distribution by Risk Tier

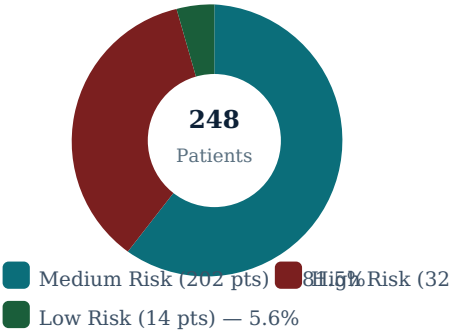


Fig 1. Patient distribution across the three pricing tiers.

Model Selection Rationale

Model	MAE	R ² Test	CV Mean R ²	Status
Linear Regression	Rs.42,029	0.651	0.819	Highest CV
XGBoost	Rs.40,590	0.667	0.745	—
Random Forest ✓	Rs.40,513	0.692	0.750	PRODUCTION

Random Forest selected as production model due to lowest individual patient prediction error (MAE Rs.40,513). All three models stable across 5-fold CV (std <0.10), confirming no overfitting.

SECTION 02

Model Intelligence & Cost Drivers

Rank	Feature	Importance	Business Implication	Pricing Action
1	ICU Stay Duration	69.6%	Single largest cost driver — each ICU day adds substantial nursing, monitoring and pharmacy cost	Tier by expected ICU days; cap at 72hr for elective Standard package
2	Implant Cost	10.4%	Devices (stents, valves) add Rs.50K–Rs.300K per case with high variance	Offer implant-included and implant-excluded package variants
3	Total Stay Length	3.6%	Overall hospitalisation captures ward + step-down costs	Include max-stay clauses in package contracts
4	Implant Flag (Y/N)	3.1%	Binary implant decision is a strong cost separator independent of implant cost value	Mandatory pre-admission implant assessment
5–6	Urea & Creatinine	2.4%	Elevated biomarkers signal organ stress → longer stay, more complications	Auto-escalate if Creatinine >2.0 or Urea >60
7+	Age, Gender, BMI	<1%	Demographics have minimal independent cost impact	Do NOT use demographics as pricing criteria

SECTION 03

Package Pricing Framework & Viable Price Points

Package prices are set at the **segment median cost + 25% financial buffer**. The buffer absorbs within-tier variance and protects the hospital from under-billing high-complexity cases while remaining competitive in the market.

STANDARD Low Risk	ADVANCED Medium Risk	COMPLEX / ICU High Risk
₹1,06,344 Recommended Package Price	₹2,10,621 Recommended Package Price	₹5,48,186 Recommended Package Price
Median Cost: ₹85,075	Median Cost: ₹1,68,497	Median Cost: ₹4,38,549
Range: ₹68,666 - ₹96,105	Range: ₹1,01,702 - ₹2,97,459	Range: ₹3,01,786 - ₹6,26,434
Buffer: +25%	Buffer: +25%	Buffer: +25%
5.6% of patients (14 pts)	81.5% of patients (202 pts)	12.9% of patients (32 pts)
✓ Elective admission ✓ No ICU required (0–1 day) ✓ No implant ✓ Normal biomarkers (Creatinine <1.2, Urea <40)	✓ 2-5 ICU days ✓ Possible implant use ✓ Semi-elective or scheduled ✓ Mild biomarker elevation (Crea 1.2–2.0, Urea 40–60)	✓ Emergency admission ✓ ICU > 6 days ✓ Implant always required ✓ Elevated Creatinine >2.0, Urea >60

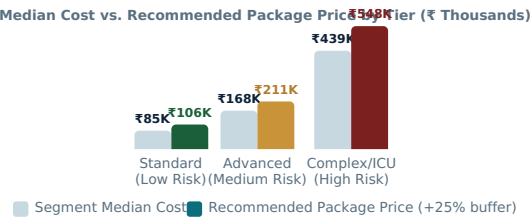


Fig 2. Median patient cost vs. recommended package price per tier. The 25% buffer above median is clearly visible in each tier.

* High-risk cases with extreme complexity (multi-valve surgery + ICU >15 days + multiple implants) should be quoted individually using the ML model output rather than a fixed package price.

SECTION 04

High-Risk Patient Segments

Risk Factor	Threshold	Avg Cost Impact	Tier Assignment	Intake Action
ICU Stay Duration	> 6 days	+Rs.1,50,000	High Risk	Assign Complex/ICU; escalate immediately
Emergency Admission	Any emergency	+Rs.60,000	High/Medium	Apply 15–20% emergency premium
Implant Required	Cost > Rs.0	+Rs.80,000	High/Medium	Mandatory pre-op implant assessment
Creatinine (renal)	> 2.0 mg/dL	+Rs.40,000	Tier upgrade	Auto-escalate; flag for clinical review
Urea (metabolic)	> 60 mg/dL	+Rs.25,000	Tier upgrade	Auto-escalate; combine with Creatinine
Total Stay Duration	> 20 days	+Rs.70,000	High Risk	Include max-stay supplemental clause

⚠ Highest-Risk Patient Profile

Emergency admission + ICU >6 days + Implant required + Creatinine >2.0. These patients average **Rs. 6,50,000+** and should always receive an individual ML-generated quote — never a standard fixed package price.

SECTION 05

Risk Management Strategies

Package pricing **transfers financial risk from patients to the hospital**. The strategies below are required to protect the hospital's financial viability across all risk scenarios — under-pricing, model drift, and clinical escalation.

● **CRITICAL** ICU Cost Overrun

ICU duration accounts for **69.6% of cost variance**. One unexpected ICU week can turn a profitable Standard package into a **Rs.2L loss**. This is the single highest financial risk in the package pricing model.

Mitigations:

- Implement ≤72hr ICU step-down protocol for all elective cases
- Add ICU-day supplemental billing clause for overruns in all package contracts
- Monitor ICU utilisation weekly per package tier

● **HIGH** Emergency Admission Premium

Emergency cases cost statistically significantly more ($p = 0.0004$). Without a structured premium, emergency admissions under package pricing will systematically generate losses.

Mitigations:

- Apply 15–20% emergency premium across all tiers
- Implement rapid triage scoring within 2 hours of admission
- Maintain separate pricing track for emergency vs. elective in all documentation

● **HIGH** Implant Cost Volatility

Implant devices range from **Rs.50K to Rs.300K** per case — a 6× spread creating enormous within-tier variance, especially in the Advanced and Complex tiers.

Mitigations:

- Negotiate bulk-purchase agreements with device suppliers with locked pricing
- Offer implant-included and implant-excluded package variants
- Mandate pre-surgical implant assessment before issuing any package price

● **MEDIUM** Model Prediction Error & Drift

The Random Forest model carries a MAE of **Rs.40,513**. On the right tail of the High Risk tier, actual errors can exceed Rs.1L. The 25% buffer absorbs most — but not all — of this uncertainty.

Mitigations:

- Monitor MAE monthly — retrain immediately if MAE rises >15% above baseline
- Quarterly mandatory retraining cadence
- Log all cases where actual cost exceeds package price for model feedback

● **MEDIUM** Biomarker Escalation Gaps

If intake staff fail to capture Creatinine and Urea at admission, patients with organ stress may be assigned the wrong pricing tier — leading to systematic cost overruns.

Mitigations:

- Make biomarker entry mandatory fields in HIS intake form
- Build auto-escalation rule: Creatinine >2.0 or Urea >60 → upgrade tier automatically
- Train intake staff on four-factor clinical scoring card

● **LOW** Small Dataset & Model Generalisation

The model was trained on 248 records — clinically acceptable but statistically limited. As patient demographics and procedures evolve, accuracy will degrade without periodic retraining.

Mitigations:

- Target 1,000+ records by Q4 through systematic data collection
- Implement model drift KPI dashboard
- Re-evaluate XGBoost vs. Random Forest when dataset exceeds 500 records

SECTION 06

Data Gaps & Model Improvement Roadmap

The current model achieves strong performance on 248 records. The following targeted data investments would materially improve pricing accuracy and enable more granular, defensible procedure-specific package catalogues.

Data Investment	Current Status	Expected Uplift	Priority
Procedure / Surgery Type (CABG, Valve, Angioplasty)	Not captured	Procedure-specific packages; R² uplift estimated +5–10%; enables specialty pricing catalogues	CRITICAL
ICD-10 Diagnosis Code	~70% missing	Specialty-specific pricing tiers; enables payer negotiations and regulatory reporting	HIGH
Insurance / Payer Type	Not captured	Net revenue modelling; negotiated rate optimisation by payer	HIGH
Pre-op Risk Score (EuroSCORE II)	Not captured	Validated clinical severity baseline; more reliable than individual biomarkers alone	HIGH
Charlson Comorbidity Index (CCI)	Partially inferred	Validated resource-use predictor; improves biomarker feature reliability	MEDIUM
Post-Discharge Outcomes	Not captured	Readmission modelling; quality-adjusted pricing; 30-day mortality tracking	MEDIUM
Surgeon / OR Team ID	Not captured	Surgeon-level efficiency analysis; OR cost attribution	MEDIUM
Pharmacy & Drug Costs	Not captured	Anticoagulants/vasopressors add 15–20% variable cost in ICU cases	LOW
Dataset Volume	248 records	Target 1,000+: XGBoost expected to outperform Random Forest at scale	ONGOING

★ **Highest-ROI Data Action**

Capturing **Procedure / Surgery Type** (CABG vs. Valve Replacement vs. Angioplasty) at admission is the single most impactful data investment. This one variable is expected to reduce package pricing error by **5–10%** and unlock procedure-specific pricing catalogues — enabling more competitive and defensible pricing to insurers and patients.

SECTION 07

Implementation Roadmap

1

Phase 1 — Immediate

Month 1

Deploy Random Forest model as internal web intake tool

Adopt three-tier pricing for all new admissions

Establish 25% buffer policy in pricing manual

Train staff on four-factor clinical scoring card

2

Phase 2 — Short-Term

Months 2–3

Implement 15–20% emergency premium; update billing

Begin capturing Procedure Type and Payer Type in EHR

Offer implant-included and implant-excluded variants

Set up monthly MAE monitoring dashboard (+15% alert)

3

Phase 3 — Medium-Term

Months 4–6

Retrain model (target 400+ records); evaluate XGBoost vs. RF

Introduce CCI and EuroSCORE II into intake scoring

Launch procedure-specific catalogues (CABG, Valve, Angioplasty)

Formalise quarterly retraining cadence in DS team OKRs

4

Phase 4 — Long-Term

Months 7–12

Scale to 1,000+ records; retrain with full feature set

Implement real-time pricing API integrated with HIS

Publish SHAP-based cost explanation reports for audits

Evaluate outpatient cardiac procedure pricing

SECTION 08

Executive Decision Summary

Should the hospital adopt package pricing?	YES — implement immediately with three-tier risk-adjusted structure	HIGH ✓
What are the viable price points?	Standard ₹1,06,344 / Advanced ₹2,10,621 / Complex ₹5,48,186 (segment median + 25% buffer)	HIGH ✓
Which patients are highest risk?	Emergency + ICU >6 days + Implant + Elevated biomarkers — 12.9% of patient volume	HIGH ✓
What risk management is required?	25% buffer + ICU step-down protocol + Emergency premium + Quarterly model retraining	HIGH ✓
What additional data is most valuable?	Procedure/Surgery Type at admission — highest expected accuracy gain (5–10% R ² uplift)	MEDIUM

This report was generated from a supervised machine learning pipeline trained on 248 cardiac patient records from Mission Hospital. The Random Forest model achieved R² = 0.692 on held-out test data and a 5-fold CV mean R² = 0.819. All pricing recommendations include a 25% financial buffer above segment median cost. This document is intended exclusively for internal management use and must not be distributed externally without prior executive approval.